Curriculum Vitae

STEVEN M. CHASE

115N Mellon Institute • Carnegie Mellon University • 4400 Fifth Avenue • Pittsburgh, PA 15213
schase@cmu.edu • (412) 268-5512
https://chaselab.github.io

EDUCATION

Ph.D.	Biomedical Engineering	Johns Hopkins University	2006
M.S.	Electrical Engineering	University of California, Berkeley	1999
B.S. *(with honors)*	Applied Physics	California Institute of Technology	1997

ACADEMIC POSITIONS

Professor — **Carnegie Mellon University** — 2021-current
Neuroscience Institute and Department of Biomedical Engineering
Courtesy Appointments: Robotics Institute, Electrical & Computer Engineering

Associate Professor — 2017-2021

Assistant Professor — 2011-2017

Visiting Research Scholar — **University of Pittsburgh** — 2010-2011
Department of Neurobiology

Post-doctoral Research Associate — **Carnegie Mellon University** — 2006-2010
Department of Statistics
Jointly mentored by:
Dr. Robert E. Kass, Dept. of Statistics, Carnegie Mellon University, and
Dr. Andrew B. Schwartz, Dept. of Neurobiology, University of Pittsburgh

HONORS AND PROFESSIONAL ACHIEVEMENTS

Elected fellow, AIMBE (American Institute of Medical and Biological Engineers)	2025
CIT Dean's Early Career Fellowship	2017
NSF CAREER Award	2016
Wimmer Faculty Fellowship	2013
Individual National Research Service Award, NIH grant DC05742	2002
California Fellowship in Microelectronics	1997
Caltech Merit Scholarship	1996

FUNDING

Current, Research

NIH (R01-NS105318) PIs: Batista, Yu, Chase Total Amount: $2,794,652
Interrogating and perturbing neural dynamics
Duration: 2/9/2026-11/30/2030

NIH (R01-NS129098) PIs: Chase, Batista, Stauffer Total Amount: $1,807,561
Neural mechanisms of motivated movement
Duration: 4/15/2023-2/29/2028

NIH (R01-NS129584) PIs: Chase, Yu, Batista Total Amount: $2,848,047
Memory formation in motor cortex
Duration: 9/21/2022-6/20/2027

NSF (DRL2124066) PIs: Chase, Yu, Smith, Batista Total Amount: $2,155,795
Collaborative Research: NCS-FR: Volitional control of internal cognitive states
Duration: 9/1/2021-9/14/2026

NIH (R01-NS121079) Co-I (PI: Collinger) CMU Amount: $244,869
Quantifying neural variability and learning during real world brain-computer interface use
Duration: 6/1/2021-11/30/2026

NIH (U01-NS123125) Co-I (PI: Collinger) Total Amount: $5,403,200
Influence of task complexity and sensory feedback on cortical control of grasp force
Duration: 6/1/2021-5/31/2026

Current, Training

NIH (R90/T90-DA060340) PIs: Chase, Torres-Oviedo Total Amount: $2,729,832
Interdisciplinary training in computational neuroscience
Duration: 9/15/2023-6/30/2028

NIH (T32-EB029365) Co-I (PI: He) Total Amount: $1,000,000
Integrative training in neural interfacing
Duration: 9/1/2021-8/31/2026

Completed

NSF (IOS1553252) - **CAREER Award** PI Total Amount: $800,000
CAREER: Neural mechanisms of skill learning
Duration: 9/1/2016-8/31/2022 (NCE)

NIH (R21-NS115036) Co-I (PI: Kuhlman) Total Amount: $403,817
Use of optical brain-computer interface to define principles of sensorimotor plasticity
Duration: 9/30/2020-5/31/2022

NIH (R01-HD071686) PIs: Chase, Yu, Batista Total Amount: $2,181,120
Shaping neural population dynamics to facilitate learning
Duration: 4/21/2017-4/20/2022

NIH (U01-NS108922) Co-I (PI: Collinger) Total Amount: $1,182,379
Context dependent processing in sensorimotor cortex

Duration: 9/1/2018-8/31/2020

NSF (BCS1533672) - **Brain-Initiative Grant** PI Total Amount: $868,950
NCS-FO: The structure of neural variability during motor learning
Duration: 9/1/2015-8/31/2020 (NCE)

PA Dept of Health Research Formula Grant PI Total Amount: $300,000
Predicting short time scale neural reorganization during learning
Duration: 1/1/2017-12/31/2019

NSF (IIS1650994) Co-I (PI: Wactler) Total Amount: $300,000
EAGER: Controlling a robotic third hand - exploring use of distributed intelligence from autonomy to brain machine interfaces for augmenting human capability
Duration: 8/15/2016-7/31/2018

Curci Foundation PIs: Chase, Gittis, Kuhlman Total Amount: $250,000
Computational and biological approaches to delineate the neural circuitry of learning
Duration: 3/1/2017-2/28/2018

NIH (R01-HD071686) Co-I (PIs: Yu & Batista) Total Amount: $1,575,000
Dissecting brain-computer interfaces: a manifold & feedback-control approach
Duration: 9/1/2011-6/1/2017 (NCE)

DARPA Co-I (PI: Collinger) Total Amount: $39,343
Neural control of dexterous manipulation and translation to a portable system (Chase-lab portion)
Duration: 9/1/2016-8/31/2017

IARPA Co-I (PI: Lee) Total Amount: $49,674
MICrONS: Visual cortical neural circuits for compositional learning and inference (Chase-lab portion)
Duration: 1/15/2016-7/15/2017

Curci Foundation PIs: Chase, Gittis, Kuhlman Total Amount: $200,000
Computational and biological approaches to delineate the neural circuitry of learning
Duration: 3/1/2016-2/28/2017

Craig H. Neilsen Foundation PIs: Chase, Yu Total Amount: $289,651
A self-recalibrating brain-computer interface
Duration: 1/1/2014-6/30/2016

Curci Foundation PIs: Chase, Gittis, Kuhlman Total Amount: $200,000
Computational and biological approaches to delineate the neural circuitry of learning
Duration: 3/1/2015-2/29/2016

CMU ProSEED Grant PIs: Chase, Gittis, Kuhlman Total Amount: $50,000
Computational and biological approaches to delineate the neural circuitry of learning
Duration: 1/1/2015-12/31/2015

DARPA PIs: Chase, Schwartz, Fedder, Cui Total Amount: $273,527
Reliable cortical interfaces (Chase-lab portion)
Duration: 3/1/2012-12/31/2012

PA Dept of Health Research Formula Grant PI Total Amount: $733,494
Correlated structure in motor cortical populations
Duration: 1/1/2012-12/31/2015

INTELLECTUAL PROPERTY

1. Disclosure (invention): Marino P, Batista AP, Yu BM, **Chase SM**, Motiwali AN, Oby ER, Smoulder AL (2023). A posture-invariant brain-computer interface. Provisional U.S. Patent.
2. Patent (pending): Oby ER, Batista AP, **Chase SM**, Degenhart AD, Golub MD, Sadtler PT, and Yu BM (2021). Method of incremental training to create new patterns of physiological control signals. Serial No. 17/343,050.
3. Disclosure (invention): Oby ER, Golub MD, Hennig JA, Degenhart AD, **Chase SM**, Yu BM, and Batista AP (2020). New neural activity patterns emerge with long-term learning. Provisional U.S. Patent Application No. 63/036,735.
4. Disclosure (invention): Yu BM, Bishop W, Degenhart A, Oby ER, **Chase SM**, and Batista AP (2018). A stabilized brain-computer interface based on neural manifold alignment. Provisional U.S. Patent Application No. 62/763,874.
5. Disclosure (source code): Golub MD, **Chase SM**, and Yu BM (2017). Matlab code for internal model estimation.
   Code publically available at (https://github.com/mattgolub/internal_model_estimation).
6. Disclosure (invention): Sharma A, Nguyen K, **Chase SM**, and Gittis A (2016). A split-wheel treadmill system for rodents. Provisional U.S. Patent Application No. 62/496,572.
7. Patent: **Chase SM**, Chang-Hasnain CJ, and Waite JM (2004). Apparatus and method for controlled cantilever motion through torsional beams and a counterweight. US Patent #6,813,053 B1

PUBLICATIONS

Preprints, Publicly Accessible

1. Smoulder AL, Marino PJ, Oby ER, Snyder SE, *Batista AP, ***Chase SM**. Reward influences movement vigor through multiple motor cortical mechanisms. boRxiv 2025.04.09.648001.
2. Fisac CF, **Chase SM.** Sensory constraints on volitional modulation of the motor cortex. bioRxiv 2023.01.22.525098

Peer Reviewed Journal Papers

3. Marino PJ, Bahureksa L, Fisac CF, Oby ER, Smoulder AL, Motiwala A, Degenhart AD, Grigsby EM, Joiner WM, ***Chase SM**, *Yu BM, *Batista AP. (2025) A posture subspace in primary motor cortex. Neuron.
4. Menendez JA, Hennig JA, Golub MD, Oby ER, Sadtler PT, Batista AP, **Chase SM**, Yu BM, Latham PE. A theory of brain-computer interface learning via low-dimensional control. eLife 14:RP106309.
5. Kunigk NG, Schone HR, Gontier C, Hockeimer W, Tortolani AF, Hatsopoulos NG, Downey JE, **Chase SM**, Boninger ML, Deklava BD, Collinger JL. Motor somatotopy impacts imagery

*** Denotes equal contribution.**

strategy success in human intracortical brain-computer interfaces. (2025) J Neural Eng 22 026004.

6. Smoulder AL, Marino PJ, Oby ER, Snyder SE, Miyata H, Pavlovsky NP, Bishop WE, Yu BM, *__Chase SM__, *Batista AP (2024). A neural basis of choking under pressure. Neuron 112:1-10.
   [Subject of a Neuron Preview and news reports in Nature, Science, and PNAS.]
7. Fitzgerald JJ, Zhou W, **Chase SM**, Joiner WM (2024). Dissociating the influence of limb posture and visual feedback shifts on the adaptation to novel movement dynamics. Neuroscience 549:24-41.
8. Losey DM, +Hennig JA, +Oby ER, Golub MD, Sadtler PT, Quick KM, Ryu SI, Tyler-Kabara EC, *Batista AP, *Yu BM, *__Chase, SM__ (2024). Learning leaves a memory trace in motor cortex. Current Bio. 34:1519-1531.
   [Subject of a Current Biology Dispatch.]
9. Deklava BM, Chowdhury RH, Batista AP, **Chase SM**, Yu BM, Boninger ML, Collinger JL (2024). Motor cortex retains and reorients neural dynamics during motor imagery. Nature Hum Behav. https://doi.org/10.1038/s41562-023-01804-5
10. Koh TH, Bishop WE, Kawahima T, Jeon BB, Srinivasan R, Kuhlman SJ, Ahrens MB, *__Chase SM__, *Yu BM (2022) Dimensionality reduction of calcium-imaged neuronal population activity. Nature Comp Sci. 3:71-85.
11. Jeon BJ, Fuchs T, **Chase SM**, Kuhlman SJ (2022) Visual experience has opposing influences on the quality of stimulus representation in adult primary visual cortex. eLife 11:e80361.
12. Jeon BB, Fuchs T, **Chase SM**, Kuhlman SJ (2022) Existing function in primary visual cortex is not perturbed by new skill acquisition of a non-matched sensory task. Nature Comm 13:3638.
13. +Smoulder AL, +Pavlovsky NP, +Marino PJ, Degenhart AD, McClain NT, *Batista AP, *__Chase SM__ (2021) Monkeys exhibit a paradoxical decrease in performance in high-stakes scenarios. PNAS 118:e2109643118.
    [Subject of a PNAS commentary.]
14. Nguyen KP, Sharma A, Gil-Silva M, *Gittis AH, *__Chase SM__ (2021) Distinct kinematic adjustments over multiple timescales accompany locomotor learning in mice. Neuroscience 466:260-272.
    [Subject of a Neuroscience commentary.]
15. Hennig JA, Oby ER, Golub MD, Bahureksa LA, Sadtler PT, Quick KM, Ryu SI, Tyler-Kabara EC, *Batista AP, *Chase SM, *Yu BM (2021) Learning is shaped by abrupt changes in neural engagement. Nature Neurosci. 24:727-736.
16. *Kowalewski N, *Kauttonen J, Stan PL, Jeon BJ, Fuchs T, **Chase SM**, Lee TS, Kuhlman SJ (2021) Development of natural scene representation in primary visual cortex requires early postnatal experience. Curr Bio. 31:369-380.
17. +Degenhart AD, +Bishop WE, Oby ER, Tyler-Kabara EC, *__Chase SM__, *Batista AP, *Yu BM (2020) Stabilization of a brain-computer interface via the alignment of low-dimensional spaces of neural activity. Nature BME 4:672-685.
    [Subject of a News & Views review.]

18. Oby ER, Golub MD, Hennig JA, Degenhart AD, Tyler-Kabara EC, *Yu BM, ***Chase SM**, *Batista AP (2019) New neural activity patterns emerge with long-term learning. PNAS. 116:15210-15215.
[Subject of a PNAS commentary.]

19. Zhou X, Tien R, Ravikumar S, and **Chase SM** (2019) Distinct types of neural reorganization during long-term learning. J Neurophysiol. 121:1329-1341.
[Received APS*select* award. Subject of a JNP Podcast.]

20. Jeon B, Swain A, Good J, **Chase SM**, and Kuhlman SJ (2018) Feature selectivity is stable in primary visual cortex across a range of spatial frequencies. Sci Rep. 8:15288.

21. Hennig JA, Golub MD, Lund PJ, Sadtler PT, Oby ER, Quick KM, Ryu SI, Tyler-Kabara EC, *Batista AP, *Yu BM, ***Chase SM** (2018) Constraints on neural redundancy. eLife 7:e36774.
[Subject of a Trends in Cognitive Sciences Spotlight. Recommended by Faculty of 1000.]

22. Downey J, Schwed N, **Chase SM**, Schwartz AB, and Collinger J (2018) Intracortical recording stability in human brain-computer interface users. J Neural Eng. 15:046016.

23. Golub MD, Sadtler PT, Oby ER, Quick KM, Ryu SI, Tyler-Kabara EC, Batista AP, ***Chase SM**, and *Yu BM (2018). Learning by neural reassociation. Nature Neurosci. 21:607-616.
[Subject of a News & Views review. Recommended by Faculty of 1000.]

24. Zhang Y and **Chase SM** (2018). Optimizing the usability of brain computer interfaces. Neural Comp. 22:1-36.

25. Bittner SR, Williamson RC, Snyder AC, Litwin-Kumar A, Doiron B, ***Chase SM**, *Smith MA, and *Yu BM (2017). Population activity structure of excitatory and inhibitory neurons. PLoS One 12(8): e0181773.

26. Suway SB, Orellana J, McMorland AJC, Fraser GW, Liu Z, Velliste M, **Chase SM**, Kass RE, and Schwartz AB (2017). Temporally segmented directionality in the motor cortex. Cerebral Cortex 7:1-14.

27. Rasmussen RG, Schwartz AB, and **Chase SM** (2017). Dynamic range adaptation in primary motor cortical populations. eLife 2017;6:e21409.

28. Oby ER, Perel S, Sadtler PT, Ruff DA, Mischel JL, Montez DF, Cohen MR, *Batista AP, and ***Chase SM** (2016). Extracellular voltage threshold settings can be tuned for optimal encoding of movement and stimulus parameters. J Neural Eng. 13:036009.

29. Golub MD, *Yu BM, and ***Chase SM** (2015). Internal models for interpreting neural population activity during sensorimotor control. eLife 10.7554/eLife.10015.
[Recommended by Faculty of 1000.]

30. Zhang Y and **Chase SM** (2015) Recasting brain-machine interface design from a physical control system perspective. Review. J Comp Neurosci. 39:107-118.

31. Perel S, Sadtler PT, Oby ER, Ryu SI, Tyler-Kabara EC, *Batista AP, and ***Chase SM** (2015). Single-unit activity, threshold crossings and local field potentials in motor cortex differentially encode reach kinematics. J Neurophysiol. 114: 1500-1512.

32. Kozai T, Du Z, Gugel Z, Smith MA, **Chase SM**, Bodily LM, Caparosa EM, Friedlander RM, Cui XT (2015) Comprehensive chronic laminar single-unit, multi-unit, and local field potential recording performance with planar single shank electrode arrays. J Neurosci Meth. 242C:15-40.

33. Sadtler PT, Quick KM, Golub MD, **Chase SM**, Ryu SI, Tyler-Kabara EC, *Yu BM, and *Batista AP (2014). Neural constraints on learning. Nature. 512:423-426.
[Cover feature and subject of a research highlight. Recommended by Faculty of 1000.]

34. Todorova S, Sadtler PT, Batista AP, **Chase SM**, and Ventura V (2014). To sort or not to sort: the impact of spike-sorting on neural decoding performance. J Neural Eng. 11:056005.

35. Golub MD, Yu BM, Schwartz AB, and **Chase SM** (2014). Motor-cortical control of movement speed with implications for brain-machine interface control. J Neurophysiol. 112:411-429.

36. **Chase SM,** Kass RE, and Schwartz AB (2012) Behavioral and neural correlates of visuomotor adaptation observed through a brain-computer interface in primary motor cortex. J Neurophysiol. 108:624-644.

37. **Chase SM,** Schwartz AB, and Kass RE (2010) Latent inputs improve estimates of neural encoding in motor cortex. J Neurosci. 30:13873-13882.

38. Legenstein R, **Chase SM**, Schwartz AB, and Maass W (2010) A reward-modulated Hebbian learning rule can explain experimentally observed network reorganization in a brain control task. J Neurosci 30:8400-8410.

39. Koyama S, **Chase SM**, Whitford A, Velliste M, Schwartz AB, and Kass RE (2010) Comparison between decoding algorithms in open-loop and closed-loop performance. J Comp Neurosci 29:73-87.

40. Legenstein R, **Chase SM**, Schwartz AB, and Maass W (2009) Functional network reorganization in motor cortex can be explained by reward-modulated Hebbian learning. In: Advances in Neural Information Processing Systems 22.

41. **Chase SM,** Schwartz AB, and Kass RE (2009) Bias, optimal linear estimation, and the differences between open-loop simulation and closed-loop performance of brain-computer interface algorithms. Neural Networks 22:1203-1213.

42. Fraser GW, **Chase SM**, Whitford A, and Schwartz AB (2009) Control of a brain-computer interface without spike sorting. J Neural Eng 6:055004.

43. *Jarosiewicz B, ***Chase SM**, Fraser GW, Velliste M, Kass RE, and Schwartz AB (2008) Functional network reorganization during learning in a brain-computer interface paradigm. PNAS 105:19486-19491.
[Recommended by Faculty of 1000.]

44. **Chase SM** and Young ED (2008) Cues for sound localization are encoded in different aspects of spike trains in the inferior colliculus. J Neurophysiol 99:1672-1682.

45. **Chase SM** and Young ED (2007) First-spike latency information in single neurons increases when referenced to population onset. PNAS 104:5175-5180.

46. **Chase SM** and Young ED (2006) Spike-timing codes enhance the representation of multiple simultaneous sound-localization cues in the inferior colliculus. J Neurosci 26:3889-3898. [Recommended by Faculty of 1000.]

47. **Chase SM** and Young ED (2005) Limited segregation of different types of sound localization information among classes of units in the inferior colliculus. J Neurosci 25:7575-7585. [Recommended by Faculty of 1000.]

Peer Reviewed Conference Papers

48. Ye J, Rizzoglio F, Smoulder A, Mao H, Ma X, Marino P, Chowdhury R, Moore D, Blumenthal G, Hockeimer W, Kunigk N, Mayo P, Batista AP, **Chase SM**, Rouse A, Boninger ML, Greenspon C, Schwartz AB, Hatsopoulos NG, Miller LE, Bouchard KE, Collinger JL, Wehbe L, Gaunt R. (*in press*) A generalist intracortical motor decoder. NeurIPS.
49. Zhang Y and **Chase SM** (2016) A control-theoretic approach to brain-computer interface design. American Control Conference (ACC) 2016:5765-5771.
50. Zhang Y and **Chase SM** (2013) A stabilized dual Kalman filter for adaptive tracking of brain-computer interface decoding parameters. Proc 35th Annual Conf IEEE EMBS 7100-7103.
51. Perel S, Sadtler PT, Godlove JM, Ryu SI, Wang W, Batista AP, and **Chase SM** (2013) Direction and speed tuning of motor-cortex multi-unit activity and local field potentials during reaching movements. Proc 35th Annual Conf IEEE EMBS 299-392.
52. Golub MD, ***Chase SM**, *Yu BM (2013) Learning an Internal Dynamics Model from Control Demonstration. JMLR Workshop & Conference Proceedings 28:606-614.
53. Golub MD, *Yu BM, and ***Chase SM** (2012) Internal models engaged by brain-computer interface control. Proc 34th Annual Conf IEEE EMBS 1327-1330.
54. Zhang Y, Schwartz AB, **Chase SM**, and Kass RE (2012) Bayesian learning in assisted brain-computer interface tasks. Proc 34th Annual Conf IEEE EMBS 2740-2743.
    [Finalist, student paper competition.]

Reviews, Book Chapters & Invited Commentaries

55. Hennig JA, Oby ER, Losey DM, *Batista AP, *Yu BM, ***Chase SM** (2021) How Learning Unfolds in the Brain: Toward an Optimization View. Neuron 109:3720-3735.
56. Oby ER, Hennig JA, Batista AP, Yu BM, **Chase SM** (2020) Intracortical brain machine interfaces. In: He B (Ed) Neural Engineering, third edition. Springer.
57. Huggins JE, Guger C, Ziat M, Zander TO, Taylor D, Tangermann M, Soria-Frisch A, Simeral J, Scherer R, Rupp R, Ruffini G, Robinson DKR, Ramsey NF, Nijholt A, Müller-Putz G, McFarland DJ, Mattia D, Lance BJ, Kindermans PJ, Iturrate I, Herff C, Gupta D, Do AH, Collinger JL, Chavarriaga R, **Chase SM**, Bleichner MG, Batista A, Anderson CW, Aarnoutse EJ (2017) Workshops of the sixth international brain-computer interface meeting: brain-computer interfaces past, present, and future. Brain-Computer Interfaces 4:3-36.
58. Golub MD, **Chase SM**, Batista AP, and Yu BM (2016) Brain-computer interfaces for dissecting cognitive processes underlying sensorimotor control. Curr Opin Neurobio 37:53-58.
59. *Yu BM and ***Chase SM** (2014) Shedding light on learning. Nature Neurosci. 17:746-747. *Commissioned commentary (News and Views) on "Voltional modulation of optically recorded calcium signals during neuroprosthetic learning" by Clancy et al., Nature Neurosci. 17:807-809.*
60. **Chase SM** (2012) Single spikes may suffice. J Neurophysiol. 108:1809-1809. *Commissioned commentary (Editorial Focus) on "Comparison of latency and rate coding for the direction of whisker deflection in the subcortical somatosensory pathway" by Storchi et al., J. Neurophysiol. 108:1810-1821.*

61. **Chase SM** and Schwartz AB (2011) Inference from populations: going beyond models. In: Enhancing performance for Action and Perception, Progress in Brain Research Volume 192:103-112 (Green AC, Chapman, E, Kalaska JF, & Lepore F, eds).

62. Guger C, Bin G, Gao X, Guo J, Hong B, Liu T, Gao S, Guan C, Ang KK, Phua KS, Wang C, Chin ZY, Zhang H, Lin R, Chua KSG, Kuah C, Ang BT, George H, Kübler A, Halder S, Hösle A, Münßinger J, Palatucci M, Pomerleau D, Hinton G, Mitchell T, Ryan DB, Sellers EW, Townsend G, **Chase SM**, Whitford AS, Schwartz AB, Kawashima K, Shindo K, Ushiba J, Liu M and Schalk G (2011) State of the art in BCI research: BCI award 2010. In: Recent Advances in Brain-Computer Interfaces (Fazel R, ed).

Reviews & Interviews About My Work

1. Gallego J (2025) Neural manifolds: more than the sum of their neurons. Nature Reviews Neuroscience.
   [Journal Club report on Sadtler et al., Nature, 2014.]

2. Adam D (2024) Brain signals show how monkeys "choke" under pressure. PNAS journal club: 10.1073/journal-club.2960026c-5f8f-4b4b-b20d-881c17c5d6d1.
   [News report on Smoulder et al., Neuron, 2024.]

3. Addaszadeh M and Ebitz B (2024) When rewards backfire: Collapsing under pressure in motor cortex. Neuron 112:3373-3375.
   [Commentary on Smoulder et al., Neuron, 2024.]

4. Wilcox C (2024) Whey do we choke under pressure? A brain imaging study in monkeys offers clues. Science https://www.science.org/content/article/why-do-we-choke-under-pressure-brain-imaging-study-monkeys-offers-clues.
   [News article on Smoulder et al., Neuron, 2024.]

5. Coleman J (2024) Why do we crumble under pressure? Science has the answer. Nature News 12 Sept 2024.
   [News article on Smoulder et al., Neuron, 2024.]

6. van der Plas M, Failla A, Robertson EM (2024) Neuroscience: Memory modification without catastrophe. Curr Bio. 34:R273-R293.
   [Commentary on Losey et al., Curr Bio., 2024.]

7. Saikley A and Haroush K (2021) Toward a neurobiological model of human performance under pressure. PNAS 118:e2113777118.
   [Commentary on Smoulder et al., PNAS, 2021.]

8. Jacobs JR and Carey MR (2021) Move over Rotarod, here comes RotaWheel. Neuroscience doi.org/10.1016/j.neuroscience.2021.05.004.
   [Commentary on Nguyen et al., Neuroscience, 2021.]

9. Wimalasena LN, Miller LE and Pandarinath C (2020) From unstable input to robust output. Nature BME 4:665-667.
   [Review of Degenhart et al., Nature BME, 2020.]

10. Pryluk R and Paz R (2019) Learning outside the box. PNAS. 116:15316-15318.
    [Commentary on Oby et al., PNAS, 2019]

11. Yates B (2019) Distinct types of neural reorganization during long-term learning. JNP Podcasts, released 22 May 2019. https://jneurophysiol.podbean.com/e/distinct-types-of-neural-reorganization-during-long-term-learning/.
[Spotlight on Zhou et al., J Neurophysiol. 2019]
12. Kalaska J (2019) F1000Prime Recommendation of [Hennig JA et al., elife 2018 7]. In F1000Prime, 22 May 2019; 10.3410/f.733821237.793560118.
13. Kalaska J (2019) F1000Prime Recommendation of [Golub MD et al., Nat Neurosci 2018 21(4):607-616]. In F1000Prime, 22 May 2019; 10.3410/f.732828264.793560117.
14. Kalaska J (2019) F1000Prime Recommendation of [Sadtler PT et al., Nature 2014 512(7515):423-426]. In F1000Prime, 22 May 2019; 10.3410/f.718546237.793560114.
15. Kao TC and Hennequin G (2018) Null ain't dull: new perspectives on motor cortex. Trends in Cognitive Sciences. 22:1069-1071.
[Spotlight on Hennig et al., eLife, 2018]
16. Galgali AR and Mante V (2018) Set in one's thoughts. Nature Neurosci 21:459-460.
[Review of Golub et al., Nature Neurosci, 2018]
17. Destexhe A and Kumar A (2017) F1000Prime Recommendation of [Golub MD et al., elife 2015 4]. In F1000Prime, 12 Apr 2017; 10.3410/f.726000531.793530347.
18. Whalley K (2014) Learning and memory: Putting limits on learning. Nature Reviews Neuroscience 15:631.
[Review of Sadtler et al., Nature, 2014]
19. Frank M (2014) F1000Prime Recommendation of [Sadtler PT et al., Nature 2014 512(7515):423-426]. In F1000Prime, 01 Sep 2014; 10.3410/f.718546237.793499190.
20. Upson, S (2014) Cyborg confidential. SA Mind 25(6):30-35.
21. Fetz E and Richardson A (2009) F1000Prime Recommendation of [Jarosiewicz B et al., Proc Natl Acad Sci USA 2008 105(49):19486-19491]. In F1000Prime, 30 Mar 2009; 10.3410/f.1158542.618726.
22. King A (2006) F1000Prime Recommendation of [Chase SM and Young ED, J Neurosci 2006 26(15):3889-3898]. In F1000Prime, 27 Apr 2006; 10.3410/f.1011522.371004.
23. Middlebrooks J (2005) F1000Prime Recommendation of [Chase SM and Young ED, J Neurosci 2005 25(33):7575-7585]. In F1000Prime, 25 Aug 2005; 10.3410/f.1027578.331641.

DISSERTATIONS AND THESES

Smoulder A (2024) Neural population mechanisms of reward-mediated performance. PhD Dissertation, Biomedical Engineering, Carnegie Mellon University.

Losey D (2023) The geometry of neural population activity during learning and memory. PhD Dissertation, Program in Neural Computation and Machine Learning, Carnegie Mellon University.

Koh TH (2023) Statistical analysis of calcium-imaged population activity. PhD Dissertation, Biomedical Engineering, Carnegie Mellon University.

Nguyen K (2022) Dissection of detailed motor behaviors and circuit function of the basal ganglia in health and disease. PhD Dissertation, Biomedical Engineering, Carnegie Mellon University.

Fisac C (2022) Sensory constraints on volitional modulation of neural activity in motor cortex. PhD Dissertation, Biomedical Engineering, Carnegie Mellon University.

Jeon B (2021) Stability and robustness of visual feature representation in mouse V1. PhD Dissertation, Biomedical Engineering, Carnegie Mellon University.

Hennig J (2021) Structure and time course of neural population activity during learning. PhD Dissertation, Neural Computation and Machine Learning, Carnegie Mellon University.

Zhou X (2020) Neural reorganization supporting long-term brain-computer interface learning. PhD Dissertation, Biomedical Engineering, Carnegie Mellon University.

Zhang Y (2018) Provably optimal design of a brain-computer interface. PhD Dissertation, Robotics Institute, Carnegie Mellon University.

Ravikumar S (2017) Tracking chronically recorded neurons using pairwise cross-correlograms. MS Thesis, Biomedical Engineering, Carnegie Mellon University.

Sharma A (2017) Design of a split-running wheel for studies of locomotor learning. MS Thesis, Biomedical Engineering, Carnegie Mellon University. (co-advised by Aryn Gittis)

Jeon B (2016) Design and testing of a closed-loop brain-computer interface system for operation by a mouse. MS Thesis, Biomedical Engineering, Carnegie Mellon University. (co-advised by Sandra Kuhlman)

Golub MD (2015) Interpreting neural population activity during feedback motor control. PhD Dissertation, Electrical and Computer Engineering, Carnegie Mellon University. (co-advised by Byron Yu)

Lund P (2014) Predicting high-dimensional neural activity patterns during brain-computer interface control. MS Thesis, Machine Learning, Carnegie Mellon University. (co-advised by Byron Yu)

Zhao Y (2014) An extended latent Kalman filter for nuisance variable removal in brain-computer interface decoding applications. MS Thesis, Biomedical Engineering, Carnegie Mellon University.

Chase SM (2006) The representation of sound localization information among neural populations in the inferior colliculus. PhD Dissertation, Biomedical Engineering, Johns Hopkins University. (Advisor: ED Young)

Chase SM (1999) A micromechanical red-shifting tunable vertical cavity filter. MS Thesis, Electrical Engineering, University of California, Berkeley. (Advisor: CJ Chang-Hasnain)

TEACHING

Classes

42-101 **Introduction to BME** (12 units)

This course provides exposure to basic biology and engineering problems associated with living systems and health care delivery. Examples are used to illustrate how basic concepts and tools of science & engineering can be brought to bear in understanding, mimicking and utilizing biological processes.

Enrollment: 74, FCE Instructor 4.2/5.0, FCE Course 4.1/5.0 (median numbers)

Taught in: Fall 2013, Spring 2015, Fall 2015, Spring 2018, Spring 2019, Spring 2020, Spring 2021, Spring 2022, Spring 2024, Spring 2025.

86-631/42-631 **Neural Data Analysis** (12 units)

This class covers the basic engineering and statistical tools in common use for analyzing neural spike train data, with an emphasis on hands-on application. Topics include neural spike train statistics, estimation theory (MLE, MAP), signal detection theory (d-prime, ROC analysis), information theory (entropy, mutual information, neural coding theories, spike-distance metrics), discrete classification (naive Bayes), and continuous decoding (PVA, OLE, Kalman).

Enrollment: 21, FCE Instructor 4.8/5.0, FCE Course 4.7/5.0 (median numbers)

Taught in: Spring 2012, Fall 2012, Fall 2013, Fall 2014, Fall 2015, Fall 2016, Fall 2017, Fall 2018, Fall 2019, Fall 2021, Fall 2025.

Guest Lectures

Carnegie Mellon University

03-161: Molecules to Mind: Spring 2015, Spring 2019, Spring 2020, Spring 2021

15-386/15-686: Neural Computation: Spring 2010, Spring 2011, Spring 2012, Spring 2014(twice)

18-101: Introduction to Electrical and Computer Engineering: Fall 2012

18-819: Neural Technology, Sensing and Stimulation: Fall 2013

36-759: Statistical Models of the Brain: Spring 2013

42-101: Introduction to BME: Spring 2012, Fall 2014, Spring 2016, Fall 2016, Spring 2017, Fall 2017, Fall 2018, Fall 2019, Fall 2020, Fall 2021, Fall 2024, Fall 2025

42-201: Professional Issues in BME: Spring 2012, Spring 2013, Spring 2015, Spring 2016, Fall 2016, Spring 2017

42-447: Rehabilitation Engineering: Fall 2017

42-630: Introduction to neuroscience for engineers: Spring 2016, Spring 2019

42-632: Neural Signal Processing: Spring 2014

42-655: Brain-Computer Interface: Principles and Applications: Spring 2022

42-690: BME in Everyday Life: Spring 2020

80-270: Philosophy of Mind: Spring 2014

85-382/18-782: The psychology and neuroscience of consciousness: Spring 2019

86-375/86-675: Computational Perception: Spring 2012

uPNC Summer Seminar Series: Summer 2011, Summer 2012, Summer 2013, Summer 2015, Summer 2016, Summer 2017, Summer 2018, Summer 2019, Summer 2020, Summer 2021, Summer 2024

Oregon Health & Science University

NEUS 627 / BEHN 640 Systems Neuroscience: Fall 2017, Fall 2018

University of Pittsburgh

BIOEN 2696: Control theory in neuroscience: Spring 2012

BIOEN 2800: Neurotechnology: Spring 2012

MSNBIO 2632: Advanced Neurophysiology: Spring 2007

PROFESSIONAL SERVICE

Current:

Co-Director: Program in Neural Computation, since 2016

Co-Director: Interdisciplinary training in computational neuroscience training program, since 2023

Member, CIT ad-hoc promotion and tenure committee, since 2024

Member, CNBC Executive Committee, since 2011

Member, CNBC Education Committee, since 2023

Member, Neuroscience Undergraduate Education Committee, since 2017

Editorial & External committee service

Editor: Neurons, Behavior, Data Analysis, and Theory, 2018 – current.

Program Committee: Translational and Computational Motor Control Workshop, 2015 – current.

Program Committee: Cosyne 2018 – 2020.

Guest editor, PLoS Comp Bio, 2015-2016.

Workshop organization

Organizer: CMU/Pitt Workshop: The Neural Basis of Internal States, Pittsburgh, PA, June 2023.

Organizer: BME track of the Machine Learning in Science and Engineering Conference, Atlanta, GA, June 2019.

Organizer: CMU/Pitt Neuro-Learning Workshop, Pittsburgh, PA, June 2018.

Session chair: Neuroinformatics session, Machine Learning in Science and Engineering 2018.

Session organizer: "Methods: How do you find a manifold, and what do you do when you've found one?" M5: Midwest Monkey Manifolds for Movement Meeting, May 2018.

BCI 2016 Workshop organizer: "Algorithms and performance using implanted devices."

BCI 2016 Workshop organizer, "Studying learning with brain-computer interfaces."

Organizer: CNBC retreat. Nov 2012 & Oct 2013.

Organizer: Biosignals Day Symposium, "Power in numbers: How many neurons are needed to…?" Feb 2012.

Session co-chair: Platform session on neural computation & statistical analysis of neural data, BMES 2009.

Review activities

Grant review: Department of Defense Peer Reviewed Medical Research Program (PRMRP), FNRS (Fonds de la Recherche Scientifique, Belgium), Israel Binational Foundation, NSF Neuronex, NSF CISS-IIS, NSF BIO-IOS, NIH LMDN, NIH SPC, Singapore Ministry of Education.

Ad-hoc reviewer for: American Controls Conference, Biological Cybernetics, Cerebral Cortex, Computational Methods and Programs in Biomedicine, Current Biology, Current Opinions in Neurobiology, eLife, Frontiers in Neuroscience, IEEE Engineering in Medicine and Biology Society, IEEE Signal Processing Letters, IEEE Transactions on Biomedical Engineering, IEEE Transactions on Neural Systems and Rehabilitation Engineering, Journal of the Association for Research in Otolaryngology, Journal of Neural Engineering, Journal of Neurophysiology, Journal of Neuroscience, Journal of Neuroscience Methods, Nature, Nature Biomedical Engineering, Nature Communications, Nature Medicine, Nature Neuroscience, Neural Computation, Neurobiology of Disease, Neuroinformatics, Neuroscience, Neuron, PLOS Computational Biology, PLOS One, PNAS, Science, Science and Translational Medicine, and Scientific Reports.

Past internal leadership & committee roles

Chair: Graduate Affairs Committee, Biomedical Engineering, 2020-2022 (member 2011-2020)

Member: SCS Dean Review Committee, 2024

Member: NI Faculty Search Committee, 2024

Member: University Committee on Faculty Promotions without Indefinite Tenure, 2019-2021.

Member: BME faculty search committee, 2018-2019.

Member: CNBC/Biological sciences faculty search committee, 2018-2019.

Member: Carnegie Mellon Neuroscience Institute planning committee, 2018.

Member: Graduate affairs committee, Program in Neural Computation, 2013-2016.

Member: BrainHub engineering search committee, 2015.

Member: BME search committee, 2014-2015.

OUTREACH

Media Mentions

9/2024 Our work demonstrating a neural basis of choking under pressure was picked up broadly by news outlets:

Coleman J (2024) The brain really does choke under pressure. Scientific American 9/16/24

Schwaiger C "Monkey study reveals science behind 'choking under pressure'." Live Science.

Sirius XM Canada, The Arlene Bynon Show

Independently written up by: medicalrepublic.com.au, psychologytoday.com, insidehook.com, newatlas.com, tecscience.tec.mx.

CMU news release picked up by: 1440 news, Medical Xpress

8/2024 Sima R (2024) Even Olympic athletes choke. Here's how we can manage the yips. Washington Post.

7/2023 Elling T "The science of why we choke under pressure." Books on pod Episode #349.

7/2023 Elling T "Why we choke under pressure." ESPN radio Austin 102.7.

5/2023 Faris N "Why do players choke under pressure? For the same reason monkeys do." theScore.

12/2022 podcast: "Neuroactivity, Brain-Computer Interfaces, and the Animal Kingdom: Carnegie Mellon Scholars' Take" on The Morning Fix by 510k Cafe

10/2021 CMU press release "Does the brain learn like a computer learns?" picked up by technologynetworks and the American Society for Engineering Education, among others.

10/2021 Radio Interview, Quirks & Quarks, CBC.

04/2021 Our work showing that monkeys choke under pressure was picked up broadly by news outlets:

Hart M, "This monkey chokes under pressure just like humans do", Nerdist. Johnson D, "At least one species of primate underperforms when the heat is on", ArsTechnica.

Also featured in news.yahoo, flipboard, dailyadvent, phys.org, and dailymail, among others.

08/2020 Quoted by Vox: Heilweil R "Elon Musk is one step closer to connecting a computer to your brain."

04/2020 Walsh B "Using brain interfaces to learn about learning", Axios.

04/2020 Press release about our work "Stabilizing brain-computer interfaces" picked up by several online news sites, including interestingengineering.com, sciencedaily.com, unite.ai, and neurosciencenews.com.

08/2019 Zhou et al. (2019) cited in an article in Axios, "Looking to AI to understand how we learn."

06/2019 Press release about our work "Pitt and CMU researchers discover how the brain changes when mastering a new skill" picked up by medindia.net, innovationtoronto.com, among others.

05/2019 Work subject of a Journal of Neurophysiology Podcast: "Distinct types of neural reorganization during long-term learning."

12/2018 Quoted by Reuters: Rapaport L "Brain implants might one day help paralyzed patients use tablets."

07/2018 FutureTech podcast: Jacobs R "Mind Control – Steven Chase and Byron Yu, Professors of Biomedical Engineering, Carnegie Mellon University – Computer brain interfaces that could allow control of body movement through thought."

04/2018 Quoted in Wareable: Caddy B "Beyond sci-fi: Building the bioic body."

03/2018 Davis K "When Learning A New Task, Our Brains Are Less Flexible Than We Thought." WESA (Pittsburgh's NPR station).

03/2018 Rennie J "Brain clings to old habits when learning new tricks." Quanta Magazine.

02/2018 Quoted in the Brown Daily Herald, "University study improves interfaces between brain, computers."

05/2017 Quoted in WIRED, "Want a true bionic limb? Good luck without machine learning."

04/2017 Contributed an 'Expert Views' piece to an article published by AXIOS, "When computers merge with our minds."

02/2016 Templeton D "CMU study explores why we make mistakes," Pittsburgh Post-Gazette.

12/2015 Press release about our work "Researchers gain understanding of why the brain makes mistakes," picked up by MedicalXpress (and shared by Neuroscience News, Science Beta, and DeepStuff.org).

07/2013 Interview, IGN (an online news service)

External Outreach

10/2024 Interviewee: Stem Pathways with Rohan Sharma. "#58 – Steven Chase: Computers and Brains at CMU"

11/2021 Speaker, "Communicating with the Brain: Improving Lives with Neuroengineering" Alumni Association in Turkey webinar

08/2018 Science Briefing, Congressman Keith Rothfus

03/2017 Judge, Pittsburgh Regional Science & Engineering Fair

10/2016 Speaker, Society of Women Engineers high school outreach day

08/2016 Speaker, Andrew Carnegie Society

02/2016 Instructor, OSHER lifelong learning course

05/2015 Science Briefing, Heinz Foundation

03/2014 Science Briefing, Congressman Chaka Fattah

03/2014 Judge, Pittsburgh Regional Science & Engineering Fair

03/2013 Science Briefing, Hillman Foundation

10/2012 Science Briefing, Hillman Foundation

Internal Outreach / Mentorship

05/2018 Panelist, NSF CAREER writing workshop
09/2017 Panelist, Young Investigator Grants information session
08/2017 Panelist, Work/Life Balance workshop
05/2017 Panelist, NSF CAREER writing workshop 04/2014Panelist, CNBC Panel, "Graduate Academic Job Sear
06/2013 Mentor, CNBC Data Blitz
05/2012 Panelist, GBMES Panel, "Discussions at the Interface of Nature & Technology"

PRESENTATIONS

Invited Talks

1. 2025. MyoTalks. "Neural mechanisms of motivated movement." *virtual*
2. 2024. Barrow Neurological Institute, Pheonix, AZ. "Cognitive factors influencing brain-computer interface control."
3. 2022. UC Davis, Davis, CA. "Dissecting learning and memory with brain-computer interfaces."
4. 2021. PMC XIII, Auckland, New Zealand. "Acting out your feelings: internal state signals in motor cortex." *virtual*
5. 2021. Ben Gurion University, Be'er Sheva, Israel. "Dissecting learning and memory with brain-computer interfaces." *virtual*
6. 2020. 6th CiNet Conference. Osaka University, Osaka, Japan. "Neural manifolds: from basic science to practical improvements in brain-computer interfaces."
7. 2019. Action Club, Northeastern, Boston, MA. "Dissecting learning processes with brain-computer interfaces."
8. 2019. Cedars-Sinai, Los Angeles, CA. "Brain-computer interfaces for dissecting motor learning."
9. 2019. Machine Learning in Science and Engineering, Altanta, GA. "A stabilized brain-computer interface based on neural manifold alignment."
10. 2019. Sungkyunkwan University, Suwon, South Korea. "Neural manifolds: from basic science to practical improvements in brain-computer interfaces."
11. 2019. IEEE 2019 Winter BCI Conference, High 1 Resort, South Korea. "Neural manifolds: from basic science to practical improvements in brain-computer interfaces."
12. 2018. Oregon Health & Science University, Portland, OR. Invited to give two classroom lectures on motor system function and brain-machine interfaces.
13. 2018. CMU/Pitt Neuro-Learning Workshop, Pittsburgh, PA. "Dissecting the neural mechanisms of learning with brain-computer interfaces."
14. 2018. Machine Learning in Science and Engineering, Pittsburgh, PA. "Using machine learning to understand biological learning."
15. 2018. Neural control of movement panel, Sante Fe, NM. "BCI learning: fast and slow."
16. 2017. Oregon Health & Science University, Portland, OR. Invited to give two classroom lectures on motor system function and brain-machine interfaces.
17. 2017. TEDxCMU Pittsburgh, PA. "The limits of learnability."

18. 2016. UPMC Enterprises, Pittsburgh, PA. "State-of-the-art in intracortical brain-computer interfaces."
19. 2016. Department of Bioengineering, University of Pittsburgh, Pittsburgh, PA. "Cognitive factors that shape neural prosthetic control."
20. 2016. Department of Biomedical Engineering, University of Rochester, Rochester, NY. "Cognitive factors in neural prosthetic control."
21. 2016. American control conference panel, Boston, MA. "A control-theoretic approach to brain-computer interface design."
22. 2015. International Neuropsychological Symposium, Collioure, France. "Internal conceptions of external systems."
23. 2015. Center for Neural Engineering Prostheses and Redwood Center for Theoretical Neuroscience, UC Berkeley, Berkeley, CA. "Internal models for interpreting neural population activity."
24. 2014. SASP-SSP Continuing Education Symposium, Pittsburgh, PA. "From BrainHub to brain-computer interfaces: new research initiatives at Carnegie Mellon."
25. 2014. Johns Hopkins Biomedical Engineering Retreat, Ocean City, MD. "From goats to brain-computer interfaces."
26. 2014. Computational Motor Control Workshop, Beer-Sheva, Israel. "Internal model mismatch is responsible for the majority of errors in neuroprosthetic control."
27. 2013. 2nd Annual Next Generation Medical Imaging Workshop, Pittsburgh, PA. "Movement-related information conveyed by non-traditional signal sources in motor cortex."
28. 2013. Fourth Annual Aspen Brain Forum, Aspen, CO. "Developing algorithms for enhanced brain-machine interfaces."
29. 2013. International BCI Meeting, Pacific Grove, CA. "Visuomotor adaptation in BCI: differences and similarities from natural reaching."
30. 2011. The IEEE Pittsburgh Section Signal Processing and EMBS Chapters, Pittsburgh, PA. "Important and unimportant factors in neural prosthetic control."
31. 2011. George Mason University, Fairfax, VA. "Visuomotor adaptation observed through a brain-computer interface."
32. 2011. Harvard University, Cambridge, MA. "Visuomotor adaptation observed through a brain-computer interface."
33. 2011. Boston University, Boston, MA. "Visuomotor adaptation observed through a brain-computer interface."
34. 2010. CNUP Annual Retreat, Wheeling, WV. "Visuomotor adaptation observed through a brain-computer interface."
35. 2010. Computational Motor Control Workshop, Beer-Sheva, Israel. "Visuomotor adaptation observed through a brain-computer interface."
36. 2010. Hebrew University, Jerusalem, Israel. "Visuomotor adaptation observed through a brain-computer interface."
37. 2010. Johns Hopkins Applied Physics Laboratory, Laurel, MD. "Important and unimportant factors in neural prosthetic control."
38. 2009. Nertworks: a BMES satellite meeting, Pittsburgh, PA. "Leveraging BCI paradigms to study neural plasticity.

39. 2009. McGowan Institute Annual Retreat, Farmington, PA. "Advances in neural prosthetic control."

Conference Presentations

2025

1. Kunigk NG, Hockeimer W, Chase SM, Boninger ML, Dekleva BD, Collinger JL (2025) Identification of a neural engagement signal in human motor cortex activity. SfN annual meeting.
2. Miyata H, Chowdhury R, Smoulder A, Zambre M, Wen J, Oby E, *Batista AP, *Chase SM (2025) Subjective preference is encoded in the motor cortex. SfN annual meeting.
3. McDonnell M, Chandrasekaran AN, Ki C, Smoulder A, Yu B, *Batista AP, *Smith MA, ***Chase SM** (2025) Choking under pressure: overcompensating for arousal? SfN annual meeting.
4. Hockeimer W, Deklava B, Kunigk NG, Boninger M, Chase SM, Collinger JL (2025) Characterization of a Stable Motor Manifold in Humans Despite Neural Variability and its Use in a Novel Intracortical BCI Decoder. SfN annual meeting.
5. Couras J, Oby E, Motiwala A, *Yu BM, ***Chase SM**, *Batista AP (2025) Remembering a reach goal shapes neural population dynamics in motor cortex. SfN annual meeting.
6. Larry N, Yu B, Batista A, *Chase SM, Smith MA (2025) Reward expectation encoding in prefrontal, premotor, and primary motor cortex is enhanced through learning. SfN annual meeting.
7. Chandrasekaran A, McDonnell M, Ki C, Yu BM, *Batista AP, *Chase SM, *Smith MA (2025) Transformations from memory to action in prefrontal and motor cortices. SfN annual meeting.
8. Miyata H, Chowdhury R, Smoulder A, Zambre M, Wen J, Oby E, *Batista AP, *Chase SM (2025) Subjective preference is encoded in the motor cortex. Simian Collective meeting.
9. Ye J, Mayo JP, Smoulder A, Ma X, Mao H, Batista A, Chase SM, Rouse A, Greenspon C, Miller LE, Hatsopoulos NG, Schwartz AB, Collinger JL, Wehbe L, Gaunt R (2025) Large scale deep network models for intracortical motor decoding and control. Neural Interfaces.
10. Smoulder AL, Marino PJ, Oby ER, Snyder SE, *Batista AP, *Chase SM (2025) Reward influences movement vigor through multiple motor cortical mechanisms. NCM annual meeting.
11. *Snyder S, *Oby E, $^{+}$Smith M, $^{+}$Batista AP, $^{+}$Chase SM, $^{+}$Yu BM (2025) A direct test of the flexibility between motor cortical areas. NCM annual meeting.
12. Hockeimer W, Dekleva B, Kunigk N, Ye J, Boninger M, Chase SM, Collinger J (2025) Chronic stability of human motor manifolds that enable iBCI cursor control. NCM annual meeting.

2024

13. Ye J, Mayo JP, Smoulder A, Ma X, Mao H, Batista A, **Chase SM**, Rouse A, Greenspon C, Miller LE, Hatsopoulos NG, Schwartz AB, Collinger JL, Wehbe L, Gaunt R (2024) Neural Data Transformer 3: A Foundation Model for Motor Cortical Decoding. SfN annual meeting.

14. Hockeimer W, Dekleva B, Kunigk N, Boninger M, **Chase SM**, Collinger J (2024) Enhancing movement-related activity in neural reach representations via intracortical BCI decoder gain perturbations. SfN annual meeting.
15. Gonzalez IC, Blumenthal GH, Dekleva BM, Okorokova EV, **Chase SM**, Boninger ML, Collinger JL (2024) Continuous offline force decoding from human M1 population activity. SfN annual meeting.
16. Kunigk NG, Schone H, Hockeimer W, Gontier C, Tortolani AF, Hastopoulos NG, Downey JE, **Chase SM**, Boninger ML, Dekleva BD, Collinger JL (2024) Motor cortex somatotopy influences control strategy success for human brain-computer interfaces. SfN annual meeting.
17. Smoulder AL, Marino PJ, Oby ER, Snyder SE, *Batista AP, ***Chase SM** (2024) Reward improves movement vigor through multiple motor cortical mechanisms. SfN annual meeting.
18. Chandrasekaran AN, McDonnell M, Ki C, Smoulder A, *Batista AP, *Yu BM, ***Chase SM**, *Smith MA (2024) Reward, perceptual difficulty, and motor difficulty drive distinct changes in cortical activity and sensorimotor behavior. SfN annual meeting.
19. McDonnell M, Chandrasekaran AN, Ki C, Smoulder A, *Yu B, *Batista AP, ***Chase SM**, *Smith MA (2024) Distinct sensory and motor components of choking under pressure SfN annual meeting.
20. Borgognon S, Smoulder AL, Miyata H, Wen J, Chandrasekaran AN, Smith MA, ***Chase SM**, *Batista AP (2024) Evidence for distinct internal states influencing the primate motor cortex. SfN annual meeting.
21. Snyder S, Oby E, *Smith MA, ***Chase SM**, *Yu BM, *Batista AP (2024) The interactions between premotor and primary motor cortex can be flexible. SfN annual meeting.
22. Zambre M, Snyder S, Oby ER, Smoulder AL, ***Chase SM**, *Schwerdt H, *Batista AP (2024) Encoding of reward in the motor cortex depends on task context. SfN annual meeting.
23. Smoulder AL, Marino PJ, Oby ER, Snyder SE, *Batista AP, ***Chase SM** (2024) Reward improves movement vigor through multiple motor cortical mechanisms. SimCo meeting.
24. McDonnell M, Chandrasekaran AN, Ki C, Smoulder A, *Yu B, *Batista AP, ***Chase SM**, *Smith MA (2024) Distinct sensory and motor components of choking under pressure SimCo meeting.
25. Chandrasekaran AN, McDonnell M, Ki C, Smoulder A, *Batista AP, *Yu BM, ***Chase SM**, *Smith MA (2024) Reward, perceptual difficulty, and motor difficulty drive distinct changes in cortical activity and sensorimotor behavior. SimCo meeting.
26. Smoulder AL, Marino PJ, Oby ER, Snyder SE, *Batista AP, ***Chase SM** (2024) Reward improves movement vigor through multiple motor cortical mechanisms. Gordon Research Conference.
27. McDonnell M, Chandrasekaran AN, Ki C, Smoulder A, *Yu B, *Batista A, *Smith MA, ***Chase SM** (2024) Distinct sensory and motor components of choking under pressure. Gordon Research Conference.
28. Blumenthal G, Deklava B, Gonzalez-Martinez J, Yu B, Batista A, Gaunt R, Boninger M, **Chase SM**, Collinger J (2024) Effect of motor and cognitive complexities on motor cortical dynamics during grasping. Brain Initiative PI meeting.

29. Ye J, Mayo JP, Rouse A, Smoulder A, Ma X, Mao H, Batista A, **Chase SM**, Greenspon C, Miller L, Hatsopoulos N, Schwartz A, Collinger JL, Wehbe L, Gaunt R (2024) Neural Data Transformer 3: A foundational model for motor decoding. NCM annual meeting.
30. Sacks J, Obe E, Hennig JA, Degenhart AD, Sadtler PT, Quick KM, Ryu SI, Tyler-Kabara EC, **Chase SM**, Yu B, Batista A, Golub MD (2024) Network models for distinguishing population-level mechanisms. Cosyne.
31. Snyder SS, Oby ER, *Smith M, ***Chase SM**, *Yu BM, *Batista AP (2024) A causal test of the interactions between brain areas. Cosyne.
32. [+]Chandrasekaran AN, [+]McDonnell M, [+]Ki C, Smoulder A, *Yu BM, *Batista AP, *Smith M, ***Chase SM** (2024) Reward and perceptual difficulty drive distinct changes in behavior and motor cortical activity. Cosyne.
33. Couras J, Oby E, Motiwala A, Snyder S, Losey D, Hennig J, *Yu BM, ***Chase SM**, *Batista AP (2024) A link between the memory trace in motor cortex and savings. Cosyne.
34. Marino PJ, Bahureksa L, Fisac C, Oby ER, Motiwala A, Grigsby E, Smoulder A, Degenhart A, Joiner WM, ***Chase SM**, *Yu BM, *Batista AP (2024) A posture subspace in primary motor cortex. Cosyne.

2023

35. Wen J, Smoulder AL, Miyata H, Borgonon S, Marino P, *Batista AP, ***Chase SM** (2023) Motor cortex separately encodes reward magnitude and reward type. SfN annual meeting.
36. Miyata H, Smoulder AL, Wen J, Borgonon S, Marino P, *Batista AP, ***Chase SM** (2023) Interactions between reward and task difficulty in M1 neural population activity. SfN annual meeting.
37. Hockeimer W, Weiss JM, Deklava BD, Weir D, Kunigk NG, **Chase SM**, Boninger ML, Collinger JL (2023) Independent use of an at-home intracortical BCI system. SfN annual meeting.
38. Couras J, Oby E, Motiwala A, Snyder S, Losey D, Hennig J, *Yu BM, ***Chase SM**, *Batista AP (2023) A link between memory traces in motor cortex and savings. SfN annual meeting.
39. Kunigk NG, Hockeimer W, Gontier C, Dekeva BD, **Chase SM**, Boninger ML, Collinger JL (2023) Pupil diameter correlates with population activity and performance in human intracortical brain-computer interface task. SfN annual meeting.
40. Snyder S, Oby E, Smith M, ***Chase SM**, *Yu BM, *Batista AP (2023) A causal test of the flexibility of interactions between cortical areas. SfN annual meeting.
41. Chandrasekaran AN, Smoulder A, McDonnell M, Ki C, Marino P, Wen J, Miyata H, Borgognon S, Yu B, *Batista A, *Smith M, ***Chase SM** (2023) Reward and task difficulty drive distinct behavioral changes in monkeys performing a challenging sensorimotor task. CMU BME Forum. [Winner, Outstanding Poster Award]
42. Wen J, Smoulder AL, Miyata H, Borgonon S, Marino P, *Batista AP, ***Chase SM** (2023) Motor cortex separately encodes reward magnitude and reward type. CMU BME Forum.
43. Miyata H, Smoulder AL, Wen J, Borgonon S, Marino P, *Batista AP, ***Chase SM** (2023) Interactions between reward and task difficulty in M1 neural population activity. CMU BME Forum.

44. Blumenthal GH, Deklava BM, Gonzalez-Martinez JA, Yu BM, Batista AP, Gaunt RA, Boninger ML, **Chase SM**, Collinger JL (2023) Distinct neural modes carry information about grasp timing and grip force in sensorimotor cortex. Brain Initiative investigators meeting.
45. Kunigk NG, Deklava BM, Liu F, **Chase SM**, Boninger ML, Collinger JL (2023) BCI performance is influenced by motor imagery strategy and somatotopic constraints. International BCI meeting.
46. Hockeimer W, Kunigk N, Deklava B, **Chase SM**, Boninger ML, Collinger JL (2023) Tracking variability in subject state and iBCI performance over time. International BCI meeting.
47. Smoulder AL, Marino PJ, Pavlovsky NP, Oby ER, Snyder SE, Bishop WE, Yu BM, ***Chase SM**, *Batista AP (2023) A neural basis of choking under pressure. CMU/Pitt Workshop: The neural basis of internal states.
48. **Chase SM** (2023) Motor memories are active in neural activity after learning. Speaker on panel: Are memories active? by Batista AP, Sternad D, Chase SM, Park SW, Swanson R, and Wolpaw J. NCM annual meeting.
49. **Chase SM** (2023) Discussant on panel: Aligned neural population dynamics provide a stable window onto motor intent. Rizzoglio, F, Ma X, O'Shea D, Chang J, **Chase SM**.
50. Smoulder AL, Marino PJ, Pavlovsky NP, Oby ER, Snyder SE, Bishop WE, Yu BM, ***Chase SM**, *Batista AP (2023) A neural basis of choking under pressure. Speaker on panel Motivated movement: how reward shapes neural activity and behavior. Smoulder AL, Jang HJ, Ahmed A, Ebitz B, and Cisek P. NCM annual meeting.
51. Smoulder AL, Marino PJ, Pavlovsky NP, Oby ER, Snyder SE, Bishop WE, Yu BM, ***Chase SM**, *Batista AP (2023) A neural basis of choking under pressure. Poster, NCM annual meeting.
52. Marino PJ, Bahureksa L, Fisac CF, Oby ER, Motiwala A, Grigsby E, Smoulder A, Degenhart A, Joiner WM, ***Chase SM**, *Yu BM, *Batista AP (2023) A "posture subspace" in motor cortex. Poster, NCM annual meeting.
53. Blumenthal GH, Deklava BM, Gonzalez-Martinez JA, Yu BM, Batista AP, Guant RA, Boninger ML, **Chase SM**, Collinger JL (2023) Distinct neural modes carry information about grasp timing and grip force in sensorimotor cortex. Poster, NCM annual meeting.
54. Deklava BM, Hockeimer WI, Kunigk NC, Chowdhury RH, Miller LE, **Chase SM**, Batista AP, Boninger ML, Collinger JL (2023) Target-based discretization of motor cortical responses during motor imagery. Poster, NCM annual meeting.
55. Marino PJ, Bahureksa L, Fisac CF, Oby ER, Motiwala A, Grigsby E, Smoulder A, Degenhart A, Joiner WM, ***Chase SM**, *Yu BM, *Batista AP (2023) A "posture subspace" in motor cortex. NCM Satellite Meeting: Computations and neural code underlying the control of posture.

<u>2022</u>

56. Smoulder AL, Marino PJ, Pavlovsky NP, Oby ER, Snyder SE, Bishop WE, Yu BM, ***Chase SM**, *Batista AP (2022) Exceptionally large rewards collapse task information in neural population activity. MLMC annual meeting.
57. Fisac CF and **Chase SM** (2022) Sensory constraints on volitional modulation of motor cortex. SfN annual meeting.

58. Kunigk NG, Dekleva BD, Herrera AJ, Liu F, **Chase SM**, Boninger ML, Collinger JL (2022) Somatotopically-congruent imagery enhances BCI control in human participants. SfN annual meeting.
59. Losey D, Hennig J, Oby ER, Golub MD, Sadtler PT, Quick KM, Ryu S, Tyler-Kabara EC, *Batista AP, *Yu BM, ***Chase SM** (2022) Learning alters neural activity to simultaneously support memory and action. SfN annual meeting.
60. Blumenthal GH, Dekleva BM, Batista AP, Boninger ML, **Chase SM**, Gaunt RA, Gonzalez-Martinez JA, Yu BM, Collinger JL (2022) Distinct neural modes are associated with static and dynamic grasping force. SfN annual meeting.
61. Smoulder AL, Marino PJ, Pavlovsky NP, Oby ER, Snyder SE, Bishop WE, Yu BM, ***Chase SM**, *Batista AP (2022) Exceptionally large rewards collapse task information in neural population activity. SfN annual meeting.
62. Marino PJ, Bahureksa L, Fisac CF, Oby ER, Motiwala A, Grigsby E, Smoulder A, Degenhart A, Joiner WM, ***Chase SM**, *Yu BM, *Batista AP (2022) Posture and motor signals are organized in primary motor cortex. SfN annual meeting.
63. Koh TH, Bishop WE, Kawashima T, Jeon BB, Srinivasan R, Mu Y, Wei Z, Kuhlman SJ, Ahrens MB, ***Chase SM**, *Yu BM (2022) Dimensionality reduction of calcium-imaged neuronal population activity. CRCNS PI Mtg.
64. Blumenthal G, Dekleva B, Batista A, Boninger M, **Chase S**, Gaunt R, Gonzalez-Martinez J, Yu B, Collinger J (2022) Distinct neural modes are associated with static and dynamic grasping force. Brain Initiative PI meeting.
65. Marino PJ, Bahureksa L, Fisac C, Oby ER, Motiwali A, Grigsby E, Smoulder A, Degenhart A, Joiner WM, ***Chase SM**, *Yu BM, *Batista AP (2022) Postural and volitional signals occupy separate neural dimensions in motor cortex. NCM annual meeting.
66. Smoulder AL, Marino PJ, Pavlovsky NP, Oby ER, Snyder SE, Bishop WE, Yu BM, ***Chase SM**, *Batista AP (2022) Exceptionally large rewards lead to a collapse in neural information about upcoming movements. Cosyne annual meeting.

2021

67. Herrera AJ, Dekleva BM, Boninger ML, Yu BM, Batista AP, **Chase SM**, Collinger JL (2021) Effect of object presence and grasp intention on M1 activity during BCI controlled reaches. Brain Initiative investigators meeting.
68. Hennig JH, Oby ER, Golub MD, Bahureksa L, Sadtler P, Quick K, Ryu S, Tyler-Kabara E, *Batista AP, ***Chase SM**, *Yu BM (2021) Learning is shaped by an abrupt change in "neural engagement". IEEE NER mini-symposium.
69. Jeon BJ, **Chase SM**, and Kuhlman SJ (2021) Integration of new skills does not perturb preexisting function in primary visual cortex. SfN Global Connectome.
70. Hennig JA, Oby ER, Golub MD, Bahureksa LA, Sadtler PT, Quick KM, Ryu SI, Tyler-Kabara EC, *Batista AP, ***Chase SM**, *Yu BM (2021) Learning is shaped by abrupt changes in neural engagement. Cosyne annual meeting.

2020

71. Herrera AJ, Deklava BM, Boninger ML, Yu BM, Batista AP, **Chase SM**, Collinger JL (2020) Effect of object presence and grasp intention on M1 activity during planning and movement execution. Human single neuron conference.
72. [+]Smoulder AS, [+]Pavlovsky NP, [+]Marino P, Degenhart AD, *Batista AP, ***Chase SM** (2020) Monkeys exhibit paradoxical decrement in performance in high-stakes scenarios – they 'choke under pressure'. MLMC workshop.
73. Fisac CF and **Chase SM** (2020) Sensory feedback impacts volitional control of single neurons even after prolonged training. CMU BME Forum.
74. [+]Pavlovsky NP, [+]Smoulder AS, Degenhart AD, Marino P, *Batista AP, ***Chase SM** (2020) Monkeys exhibit paradoxical decrement in performance in high-stakes scenarios. CMU BME Forum.
75. Losey DM, Hennig JA, Oby ER, Golub MD, Sadtler PT, Quick KM, Ryu S, Tyler-Kabara EC, *Batista AP, *Yu BM, ***Chase SM** (2020) Evidence of a memory trace in motor cortex after short term learning. Cosyne annual meeting.
76. Koh TH, Bishop W, Kawashima M, Ahrens M, ***Chase SM**, *Yu BM (2020) Simultaneous dimensionality reduction and deconvolution for calcium imaging recordings. Cosyne annual meeting.
77. Menendez J, Hennig J, Golub M, Oby E, Batista A, **Chase SM**, Yu BM, Latham P (2020) A motor cortical model of brain-machine interface learning, fast and slow. Cosyne annual meeting.

2019

78. Losey DM, Hennig JA, Oby ER, Golub MD, Sadtler PT, Quick KM, Ryu S, Tyler-Kabara EC, *Batista AP, *Yu BM, ***Chase SM** (2019) Evidence of a memory trace in motor cortex after short term learning. SfN annual meeting.
79. Pavlovsky N, Degenhart AD, Marino P, *Batista AP, ***Chase SM** (2019) Monkeys choke under pressure. SfN annual meeting.
80. Jeon BB, ***Chase SM**, *Kuhlman SJ (2019) Use of optical brain computer interface to identify neural strategies underlying new skill acquisition and performance following extended practice. SfN annual meeting.
81. Herrera AJ, Gaunt RA, Boninger ML, Batista AP, Yu BM, **Chase SM**, Collinger JL (2019) The influence of object presence on M1 activity during movement planning and execution. SfN annual meeting.
82. Fitzgerald J, Zhou W, **Chase SM**, Joiner WM (2019) Sensory dissociation reveals postural influences on motor adaptation. SfN annual meeting.
83. [+]Stan PL, [+]Kauttonen J, Jeon BB, Fuchs T, **Chase SM**, Lee TS, Kuhlman SJ (2019) Impact of standard experience on tuning diversity and natural scene discrimination in primary visual cortex. SfN annual meeting.
84. Bahureksa L, Joiner WM, **Chase SM** (2019) Influence of movement workspace on M1 neural representations of arm movement. CMU BME Forum.
85. Koh TH, Bishop WE, ***Chase SM**, *Yu BM (2019) Simultaneous dimensionality reduction and deconvolution of calcium imaging activity. CMU BME Forum.

86. Jeon BB, *__Chase SM__, *Kuhlman SJ (2019) Well-trained mice maintain distinct but multiple neural strategies to solve a task. CNBC Retreat.

87. Koh TH, Bishop WE, *__Chase SM__, *Yu BM (2019) Simultaneous dimensionality reduction and deconvolution of calcium imaging activity. SAND 9 Workshop.

88. Jeon BB, **Chase SM**, Kuhlman SJ (2019) Feature selectivity is stable in primary visual cortex across a range of spatial frequencies. SAND 9 Workshop.

89. Fisac CF and **Chase SM** (2019) Sensory context interferes with volitional modulation of single neurons. NCM annual meeting.

90. Pavlovsky N, Degenhart A, Marino P, *Batista A, *__Chase SM__ (2019) Monkeys choke under pressure. Brain initiative PI meeting.

91. Collinger JL, Batista AP, Boninger ML, Herrera AJ, **Chase SM**, Gaunt RA, Karapetyan V, Loughlin P, Tyler-Kabara E, Yu BM (2019) Context-dependent processing in sensorimotor cortex. Brain Initiative PI meeting.

92. Herrera AJ, Downey JE, Gaunt RA, Boninger ML, Batista AP, Yu BM, **Chase SM**, Collinger JL (2019) Object presence changes latent neural activity in motor cortex during reaching. Brain Initiative PI meeting.

2018

93. Nguyen KP, Sharma A, Liu J, Gil-Silva MR, Maheshwari N, *Gittis AH, *__Chase SM__ (2018) Mice learn to modulate intra-and inter-limb paw kinematics with training on a novel locomotor behavioral paradigm. SfN annual meeting.

94. Herrera AJ, Degenhart AD, Bishop WE, Oby ER, Tyler-Kabara EC, **Chase SM**, Batista AP, Yu BM, Collinger JL (2018) Neural signal stabilization improves human intracortical BCI control. SfN annual meeting.

95. Whitford AD, **Chase SM**, Schwartz AB (2018) Patterns of cortical population activity during intentional control of single neurons. SfN annual meeting.

96. Fisac CF and **Chase SM** (2018) Dissociating volitional and non-volitional drivers of activity in M1. CMU BME Forum.

97. Nguyen KP, Sharma A, Liu J, Gil-Silva MR, Maheshwari N, *Gittis AH, *__Chase SM__ (2018). Mice learn to modulate intra-and inter-limb paw kinematics with training on a novel locomotor behavioral paradigm. CMU BME Forum.
[Winner, best poster award.]

98. Bahureksa L, Joiner W, **Chase SM** (2018) Comparing M1 neural reorganization during contralateral and ipsilateral visuomotor rotation learning. CMU/Pitt Neuro-Learning Workshop.

99. [+]Degenhart A, [+]Bishop W, Oby E, Tyler-Kabara E, *__Chase SM__, *Batista AP, *Yu BM (2018) A stabilized brain-computer interface based on neural manifold alignment. CMU/Pitt Neuro-Learning Workshop.

100. Oby E, Golub M, Hennig J, Degenhart A, Tyler-Kabara E, *Yu B, *__Chase SM__, *Batista A (2018). Long term learning can change neural covariability. CMU/Pitt Neuro-Learning Workshop.

101. Nguyen KP, Sharma A, Liu J, Gil-Silva MR, Maheshwari N, *Gittis AH, ***Chase SM** (2018). Mice learn to modulate intra-and inter-limb paw kinematics with training on a novel locomotor behavioral paradigm. CMU/Pitt Neuro-Learning Workshop.

102. Bishop W, Crowder E, Zandvakili A, Zhou X, **Chase SM**, Kohn A, Olson C, and Yu B (2018). Leveraging low-dimensional structure in neural population activity to combine neural recordings. CMU/Pitt Neuro-Learning Workshop.

103. Herrera AJ, Downey JE, Moran DE, Batista AP, **Chase SM**, Yu BM, Boninger ML, Gaunt RA, Collinger JL (2018) Population-level changes in primary motor cortex induced by the presence of an object. BCI 2018.

104. **Chase SM** and Collinger JL (2018) Practical uses of manifolds. M5: Midwest Monkey Manifolds for Movement Meeting.

105. **Chase SM** (2018) BCI learning: fast and slow. Part of a panel: Torres-Oviedo G, Haith A, Penhune V, **Chase SM** (2018) The role of practice in motor learning: from sensorimotor adaptation to mastering de-novo skills. NCM annual meeting.

106. Jeon BB, **Chase SM**, Kuhlman SJ (2018) Feature selectivity is stable in primary visual cortex across a range of spatial frequencies. Cold Spring Harbor Neural Circuits meeting.

107. Oby E, Golub M, Hennig J, Degenhart A, Tyler-Kabara E, *Yu B, ***Chase SM**, *Batista A (2018). Learning can change neural covariability. Brain initiative PI meeting.

108. Oby E, Golub M, Hennig J, Degenhart A, Tyler-Kabara E, *Yu B, ***Chase SM**, *Batista A (2018). Learning can generate new patterns of neural population activity. Cosyne annual meeting. [Accepted as a talk: 4% acceptance rate.]

109. Bishop W, Crowder E, Zandvakili A, Zhou X, **Chase SM**, Kohn A, Olson C, and Yu B (2018). Leveraging low-dimensional structure in neural population activity to combine neural recordings. Cosyne annual meeting.

<u>2017</u>

110. Bahureksa L, Joiner W, **Chase SM** (2017) Comparing M1 neural reorganization during contralateral and ipsilateral visuomotor rotation learning. SfN annual meeting.

111. Zhou W, Colucci-Chang K, **Chase SM**, Joiner WM (2017) Dissociating the influence of postural and visual shifts on the transfer of motor adaptation to novel workspace locations. SfN annual meeting.

112. Jeon B, Quick K, **Chase SM**, Kuhlman SF (2017) Stability of orientation and spatial frequency tuning in mouse primary visual cortex. SfN annual meeting.

113. Stan P, Kauttonen J, Jeon B, **Chase SM**, Lee TS, Kuhlman SF (2017) Primary visual cortex encodes orientation-invariant attributes of complex images. SfN annual meeting.

114. $^{+}$Bishop W, $^{+}$Degenhart A, Oby ER, Tyler-Kabara EC, Batista AP, ***Chase SM**, *Yu BM (2017) Extracting stable representations of neural population state from unstable neural recordings. SAND 8 Workshop.

115. Joiner WM, Colucci-Chang K, Zhou W, and **Chase SM** (2017) Dissociating the influence of postural and visual shifts on the transfer of motor adaptation to novel workspace locations. NCM annual meeting.

116. Hennig JA, Golub MD, Lund PJ, Sadtler PT, Quick KM, Ryu SI, Tyler-Kabara EC, Batista AP, *Yu BM, ***Chase SM** (2017) Predicting neural activity in behaviorally irrelevant dimensions. Cosyne annual meeting.

117. [+]Bishop W, [+]Degenhart A, Oby ER, Tyler-Kabara EC, ***Chase SM**, *Batista AP, *Yu BM (2017) Extracting stable representations of neural population state from unstable neural recordings. Cosyne annual meeting.

118. Golub MD, Sadtler PT, Quick KM, Ryu SI, Tyler-Kabara EC, Batista AP, ***Chase SM**, *Yu BM (2017) Learning by neural reassociation. Cosyne annual meeting.

2016

119. Golub MD, Sadtler PT, Quick KM, Ryu SI, Tyler-Kabara EC, Batista AP, ***Chase SM**, *Yu BM (2016) Learning by neural reassociation. Brain-Initiative PI meeting.

120. Flesher SN, Downey JE, Weiss J, Tyler-Kabara EC, **Chase SM**, Schwartz AB, Boninger ML, Collinger JL, Gaunt RA (2016) Grasp Force Control with Somatosensory Feedback in a Brain–Computer Interface. Brain-Initiative PI meeting.

121. Hennig JA, Golub MD, Lund PJ, Sadtler PT, Quick KM, Ryu SI, Tyler-Kabara EC, Batista AP, *Yu BM, ***Chase SM** (2016). Predicting neural activity in behaviorally irrelevant dimensions. SfN annual meeting.

122. Golub MD, Sadtler PT, Quick KM, Ryu SI, Tyler-Kabara EC, Batista AP, ***Chase SM**, *Yu BM (2016) Population-level changes in neural activity during learning. SfN annual meeting.

123. [+]Degenhart A, [+]Bishop W, Oby ER, Tyler-Kabara EC, Batista AP, ***Chase SM**, *Yu BM (2016). Self-recalibrating brain-computer interfaces based on population subspace alignment. SfN annual meeting.

124. Oby ER, Degenhart A, Tyler-Kabara E, Yu B, **Chase SM**, and Batista AP (2016) Learning to generate new patterns of neural activity. Brain Day, University of Pittsburgh.

125. [+]Degenhart A, [+]Bishop W, Oby ER, Tyler-Kabara EC, Batista AP, ***Chase SM**, *Yu BM (2016). Self-recalibrating brain-computer interfaces based on population subspace alignment. Brain Day, University of Pittsburgh.

126. Zhang Y and **Chase SM** (2016). A control-theoretic approach to brain-computer interface design. American Controls Conference 2016.

127. **Chase SM** (2016) Provably optimal design of intracortical BCI decoding algorithms. BCI 2016 workshop, "Algorithms and performance using implanted devices" organized by Chase, Batista, Yu, and Carmena.

128. Zhou X, Tien R, and **Chase SM** (2016) Distinct timescales of cortical reorganization in a long-term learning task. BCI 2016.

129. Golub MD, **Chase SM**, Batista AP, and Yu BM (2016) Brain-computer interfaces for dissecting cognitive processes underlying sensorimotor control. Complexity of Neural Computation and Cognition Workshop, Janelia Farms.

130. Oby ER, Degenhart A, Tyler-Kabara E, Yu B, **Chase SM**, and Batista AP (2016) Exploring learning-induced changes in neural population activity. NCM annual meeting.

131. Zhou X, Tien R, and **Chase SM** (2016) Distinct timescales of cortical reorganization in a long-term learning task. Cosyne annual meeting.

132. Bittner S, Williamson RC, Snyder AC, Litwin-Kumar A, Doiron B, *Smith MA, and *Yu BM (2016) Effects of excitatory versus inhibitory neuron sampling on outputs of dimensionality reduction. Cosyne annual meeting.

133. Golub MD, Sadtler PT, Quick KM, Ryu SI, Tyler-Kabara EC, Batista AP, ***Chase SM**, and *Yu BM (2016) Learning engages both high- and low-covariance modes of neural population activity. Cosyne annual meeting.

134. Golub MD, Sadtler PT, Quick KM, Ryu SI, Tyler-Kabara EC, Batista AP, ***Chase SM**, and *Yu BM (2016) Population-level changes in neural activity during learning. Cosyne workshop, "Sensorimotor learning through multi-dimensional spaces" organized by Golub.

2015

135. Oby ER, ***Chase SM**, *Yu BM, *Batista AP (2015) Exploring learning-induced changes in neural manifolds. Brain Initiative PI meeting.

136. Zhou X, Tien R, and **Chase SM** (2015) Distinct timescales of cortical reorganization in a long-term learning task. TCMC workshop.

137. Alemayehu B, Pavlovsky N, Chiou J, **Chase SM**, and Batista AP (2015) Eye-centered tuning is weak in dorsal premotor cortex when monkeys are not trained to fixate. SfN annual meeting.

138. Zhou X, Tien R, and **Chase SM** (2015) Distinct timescales of cortical reorganization in a long-term learning task. SfN annual meeting.

139. Zhang Y and **Chase SM** (2015) Recasting brain-machine interface design from a physical control system perspective. SAND 7 Workshop.

140. Matano F, **Chase SM**, Shalizi C, and Ventura V (2015). Improved kinematic models for decoding neural activity. SAND 7 Workshop.

141. Golub MD, *Yu BM, and ***Chase SM** (2015) Internal models for interpreting neural population activity during sensorimotor control. McGovern Institute Spring Symposium.

142. Golub MD, *Yu BM, and ***Chase SM** (2015) Internal models for interpreting neural population activity during sensorimotor control. Cosyne annual meeting.

2014

143. Kappel D, Pecevski D, Whitford AS, **Chase SM**, Maass W (2014). Stochastic motor control as probabilistic inference in spiking neural networks with noise. SfN annual meeting.

144. Rasmussen RG, **Chase SM**, Schwartz AB (2014). Limits on transmission of information in primary motor cortex during multidimensional reaches. SfN annual meeting.

145. Suway SB, McMorland AJC, Fraser GW, Sohn JW, **Chase SM**, Liu Z, Velliste M, Kass RE, Schwartz AB (2014) Short time-scale stability of directional tuning in motor cortex measured using maximum likelihood estimation. SfN annual meeting.

146. Rasmussen RG, Swartz AB, and **Chase SM** (2014). Dynamic range adaptation in motor cortical neurons. Cosyne annual meeting.

2013

147. Perel S, Sadtler PT, Ryu SI, Batista AP, **Chase SM** (2013). Tuning to reach kinematics of single-unit activity, threshold-crossings, and local field potentials in motor cortex. SfN annual meeting.

148. Alemayehu B, Pavlovsky N, Tyler-Kabara EC, **Chase SM**, Batista A (2013). Neurons in monkey dorsal premotor cortex are weakly sensitive to eye position when gaze fixation is untrained. SfN annual meeting.

149. Zhang Y and **Chase SM** (2013) A stabilized dual Kalman filter for adaptive tracking of brain-computer interface decoding parameters. EMBC.

150. Perel S, Sadtler PT, Godlove JM, Ryu SI, Wang W, Batista AP, **Chase SM** (2013) Direction and speed tuning of motor-cortex multi-unit activity and local field potentials during reaching movements. EMBC.

2012

151. Todorova S, **Chase SM,** Ventura V (2012). Simple automatic spike sorting can improve the efficiency of decoding from neuron spike trains. WiML 2012, Lake Tahoe, NV.

152. Todorova S, **Chase SM,** Ventura V (2012). Model-Based Clustering of non-Poisson, non-homogenous Point Processes Events with Application to Neuroscience. 7th Annual Machine Learning Workshop of the NYAS. New York, NY.

153. Golub MD, *Yu BM, and *__Chase SM__ (2012) Internal models engaged by brain-computer interface control. SfN annual meeting.

154. Zhang Y, Schwartz AB, **Chase SM**, and Kass RE (2012) Bayesian learning in assisted brain-computer interface tasks. EMBC.

155. Golub MD, *Yu BM, and *__Chase SM__ (2012) Internal models engaged by brain-computer interface control. EMBC.

156. Golub MD, Yu BM, Schwartz AB, and **Chase SM** (2012) Enhanced stability of cursor stopping in brain-computer interfaces. EMBC.

157. Todorova S, **Chase SM**, and Ventura V (2012) Model-Based Clustering of non-Poisson, non-homogenous Point Processes Events with Application to Neuroscience. Classification Society Annual Meeting.

158. Golub MD, *__Chase SM__, and *Yu BM (2012) Internal model estimation for closed-loop brain-computer interfaces. Young Investigator Talk, SAND 6 Workshop.

159. Todorova S, **Chase SM**, and Ventura V (2012) Simple automatic spike sorting methods improve decoding accuracy in a 3D movement task. SAND 6 Workshop.

160. Zhang Y, Schwartz AB, **Chase SM**, and Kass RE (2012) Bayesian learning in assisted brain-computer interface tasks. SAND 6 Workshop.

161. Golub MD, *__Chase SM__, and *Yu BM (2012) Internal model estimation for feedback control in brain-computer interfaces. Cosyne annual meeting & Machine Learning, Statistical Inference, and Neuroscience, Janelia Farms.

2011

162. Rao H, Schwartz AB, and **Chase SM** (2011) Adaptation to visuomotor gains applied through a brain-computer interface. SfN annual meeting.

163. Golub MD, Yu BM,Schwartz AB, and **Chase SM** (2011) Improving cursor stops in closed-loop brain-computer interfaces by leveraging trajectory curvature. SfN annual meeting.

2010

164. **Chase SM**, Whitford AS, Kass RE, and Schwartz AB (2010) Operant conditioning to identify volitionally controllable patterns of neural activity. SfN annual meeting.

165. Whitford AS, **Chase SM**, and Schwartz AB (2010) Coordination and control of cortical activity patterns identified via operant conditioning. SfN annual meeting.

166. McMorland AJC, Liu Z, **Chase SM**, Velliste M, Kass RE, and Schwartz AB (2010) Towards better prediction of temporal dynamics of M1 firing rates during reaching. SfN annual meeting.

167. **Chase SM**, Kass RE, and Schwartz AB (2010) Differentiating global and local adaptation responses to visuomotor rotations of a brain-computer interface. CNS10 Workshop.

168. **Chase SM** (2010) Separating intention from action in motor cortical tuning curves. Young Investigator Talk, SAND 5 Workshop.

2009

169. Legenstein R, **Chase SM**, Schwartz AB, and Maass W (2009) A reward-modulated Hebbian learning rule can explain experimentally observed network reorganization in a brain control task. NIPS

170. **Chase SM**, Schwartz AB, and Kass RE (2009) Adaptive gain modulation in the motor system observed through a brain-computer interface. SfN annual meeting.

171. Whitford AS, **Chase SM**, and Schwartz AB (2009) Task-specific tuning properties of motor cortical units used to control a brain-computer interface. SfN annual meeting.

172. Fraser GW, **Chase SM**, Koyama S, Kass RE, and Schwartz AB (2009) Important and unimportant factors for neural prosthetic control. SfN annual meeting.

173. **Chase SM**, Schwartz AB, and Kass RE (2009) Computational approaches for investigating network adaptive behavior in a brain-computer interface. BMES annual meeting.

174. **Chase SM**, Kass RE, and Schwartz AB (2009) Recent advances in neural prosthetic control. BMES annual meeting.

175. Schwartz AB, Kass RE, and **Chase SM** (2009) Learning and decoding in a multielectrode brain-machine interface paradigm. CRCNS PI meeting.

176. Koyama S, **Chase SM**, Whitford A, Velliste M, Schwartz AB, and Kass RE (2009) Comparison between decoding algorithms in open-loop and closed-loop performance. Cosyne annual meeting.

2008

177. **Chase SM**, Kass RE, and Schwartz AB (2008) A framework for modeling the closed-loop performance of brain-computer interface algorithms. SfN annual meeting.

178. Legenstein R, **Chase SM**, Schwartz AB, and Maass W (2008) A model for learning effects in motor cortex that may facilitate the brain control of neuroprosthetic devices. SfN annual meeting.

179. Koyama S, Whitford A, **Chase SM**, Velliste M, Schwartz AB, and Kass RE (2008) Recursive Bayesian algorithms for real-time cortical control of 3D neuroprosthetic devices. SfN annual meeting.

180. **Chase SM**, Schwartz AB, and Kass RE (2008) General adaptation strategies used to compensate for perturbations of a brain-computer interface. SAND 4 Workshop.

pre-2008

181. **Chase SM**, Fraser GW, Kass RE, and Schwartz AB (2007) Neural adaptation to a 3-dimensional, rotation perturbation in a closed loop brain-computer interface. SfN annual meeting.

182. **Chase SM**, Kass RE, and Schwartz AB (2007) BCI – A novel paradigm for studies of neural plasticity. BMES annual meeting.

183. **Chase SM**, Fraser GW, Kass RE, and Schwartz AB (2007) Neural adaptation to a 2-dimensional rotation perturbation in a closed loop brain-computer interface. CRCNS PI meeting.

184. **Chase SM** and Young ED (2006) Using a population reference for stimulus onset time in first spike latency coding. Cosyne annual meeting.

185. **Chase SM** and Young ED (2005) Spike timing dependent information on sound location in the inferior colliculus. SfN annual meeting.

186. **Chase SM** and Young ED (2005) The representation of sound location information among neural populations in the inferior colliculus ARO 28th midwinter meeting.

187. **Chase SM** and Young ED (2004) The processing of sound location cues by classes of units in the inferior colliculus. ARO 27th midwinter meeting.